

Project Rosetta: The Approximation Entropy & The Fractal Limits of Digital Physics

Quantifying the Irreducible Cost of Translating Continuous
Physical Dynamics into Discrete Digital Computation

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Abstract

The superconducting transmon is not a bad qubit. It is a **qudit** – a multi-level quantum digit with approximately 12 resolvable energy levels – that the quantum computing industry has spent two decades forcing into a binary straitjacket. Planck’s quantization of the electromagnetic field gives every bosonic mode a discrete, unbounded integer ladder ($E_n = n\hbar\omega$). The field’s decision to truncate this ladder to $n \in \{0, 1\}$ is not a physics constraint. It is a **radix error**: the imposition of base-2 (binary) onto a system that naturally carries base- d (where $d \approx 12$ for state-of-the-art transmons). We quantify the irreducible thermodynamic cost of this binary truncation as $S_{\text{Base}} = \ln(d/2) \approx 1.79$ nats per state projection, and recast the full approximation entropy framework as a decomposition across four cognitive and physical truncation layers. We prove that the translation functor $F : \mathcal{P} \rightarrow \mathcal{C}$ remains non-exact under the corrected framing (Theorem 1), and we propose a constructive alternative: replace qubit quantum error correction with qudit QEC on the same hardware. The transmon was never a qubit. Planck knew this in 1900. It is time the field caught up.

1. Introduction

1.1 The Binary Straitjacket

The global quantum computing industry has absorbed approximately 35 billion dollars in combined public and private investment over two decades while delivering zero commercially viable machines (QNFO Research Collective 2026e). This paper argues that the failure is not primarily an engineering problem but a **radix error** – the systematic imposition of binary logic (base-2) onto a physical system that naturally supports multi-level quantum digits (base- d).

The transmon is a **qudit**, not a qubit. The two-level truncation is a human cognitive artifact – binary is convenient for classical engineers with ten fingers – not a physical constraint. Planck’s 1900 quantization of the electromagnetic field gave us a discrete, equally-spaced energy ladder ($E_n = n\hbar\omega$, $n = 0, 1, 2, 3, \dots$). The ladder is real. It is discrete. It is also unbounded. The error is not that we discretized. The error is that we stopped at two rungs.

The key distinction (which V1.0 of this framework mischaracterized as “continuous versus discrete”):

	Old Framing (V1.0)	Corrected Framing (V4.0)
Transmon spectrum	“Continuous bosonic dynamics”	Discrete integer ladder (Planck quantization)
The field’s error	Treating analog as digital	Treating base- d as base-2
The cost	Translating continuous $\rightarrow$ discrete	Truncating base- $d \rightarrow$ base-2: $S_{\text{Base}} = \ln(d/2)$
What is lost	Symplectic capacity (abstract)	Information in rungs $n \geq 2$ (concrete, measurable as leakage)
Constructive proposal	“Three paths forward” (reactive)	Build qudits on existing hardware (constructive)

Planck gave us a digital system. We chose to read it in binary. The cost of that choice – the entropy of the binary truncation – is the subject of this paper.

1.2 The Planck-Radix Correction (Informal)

A bosonic mode in a cavity has energy eigenvalues $E_n = n\hbar\omega$ for $n = 0, 1, 2, 3, \dots$. This spectrum is:

1. **Discrete** (Planck): The eigenvalues are isolated points, not a continuum. Boundary conditions and the quantum of action $\hbar$ guarantee this.
2. **Unbounded** (Bose statistics): There is no Pauli exclusion cap on n . Coherent states routinely populate $n \sim 10^{16}$ photons in a single mode.
3. **Equally spaced** (Harmonic): $E_{n+1} - E_n = \hbar\omega$ for all n .

The transmon inherits this structure. Its Hamiltonian $H = 4E_C n^2 - E_J \cos(\phi)$ approximates a harmonic oscillator with anharmonicity $\alpha_r = \sqrt{8E_C/E_J} - 1 \approx 1.9\%$. The spectrum is approximately:

$$E_n \approx \sqrt{8E_J E_C} \left(n + \frac{1}{2} \right) - \frac{E_C}{12} (6n^2 + 6n + 3)$$

This is a discrete, unbounded, gently anharmonic ladder. Approximately 12 levels are experimentally resolvable (Koch et al. 2007; Wang et al. 2022). The field calls this a “qubit” by imposing:

$$|\psi_q\rangle = c_0|0\rangle + c_1|1\rangle$$

with projection operator $P_{\text{qubit}} = |0\rangle\langle 0| + |1\rangle\langle 1|$. The Planck Loss – the fraction of state norm discarded by this truncation – is:

$$\mathcal{L}_P = 1 - \text{Tr}(P_{\text{qubit}}|\psi\rangle\langle\psi|) = \sum_{n=2}^{\infty} |c_n|^2$$

For a coherent drive (a gate operation), $\mathcal{L}_P$ is substantial – leakage into $n \geq 2$ is the dominant error mechanism in transmon processors (QNFO Research Collective 2026f).

The radix error, stated precisely: Planck’s quantization outputs a unary/integer basis ($n = 0, 1, 2, 3, \dots$). The field forces this into binary ($n \in \{0, 1\}$) by discarding $n \geq 2$. The entropy cost is $S_{\text{Base}} = \ln(d/2)$ where d is the number of resolvable levels. For $d = 12$: $S_{\text{Base}} \approx 1.79$ nats ≈ 2.58 bits per qubit.

1.3 Statement of Results

Axis	Core Result
1. Categorical Foundations	The functor $F : \mathcal{P} \rightarrow \mathcal{C}$ is non-exact (Theorem 1) – the radix error is formal, not contingent
2. Algebraization Complexity	$\cos(\phi)$ requires degree-224 polynomial for spectral fidelity 10^{-4}
3. Thermodynamics of Translation	Rosetta’s Constant $R = k_B T \ln(N_{\text{max}}/d_{\text{comp}})$
4. Complexity Bounds	Digital Speed Limit: depth grows exponentially in qubit count Q
5. Experimental Protocol	Translation Calorimeter – falsifiable measurement of S_{Base}
HAL. Hidden Axioms	$S_{\text{Total}} = S_{\text{Obs}} + S_{\text{Base}} + S_{\text{Arch}} + S_{\text{Alg}} + S_{\text{Trot}}$ – five layers of truncation entropy

1.4 Prior Art

This work builds on and extends several independent lines of inquiry. Planck (1900) established the discrete bosonic ladder structure. De Gosson (2012) framed the conflict between geometry and algebra in quantum information using Gromov’s non-squeezing theorem, but stopped at the existence argument without quantification. The QNFO Research Collective’s “Qubit Delusion” series (QNFO Research Collective 2026e, 2026a, 2026c, 2026d, 2026b) provided the philosophical scaffolding. Wang et al. (2022) experimentally confirmed $d \approx 12$ resolvable transmon levels. Our contribution is the **radix correction** to the V1.0 thesis and the constructive qudit alternative.

2. Categorical Foundations

2.1 Category $\mathcal{P}$: Physical Dynamics

$\mathcal{P}$ has as objects **symplectic manifolds** (M^{2n}, ω) . Morphisms are Hamiltonian flows $\varphi_t^H : M \rightarrow M$ generated by $H : M \rightarrow \mathbb{R}$, or unitarily via $U(t) = e^{-iHt/\hbar}$. For the transmon: phase space is the cylinder $M = S^1 \times \mathbb{R}$ with $H = 4E_C n^2 - E_J \cos(\phi)$ (Koch et al. 2007).

2.2 Category $\mathcal{C}$: Computational Logic

$\mathcal{C}$ has as objects finite-dimensional state spaces $\mathcal{H}_2^{\otimes n}$. Morphisms are finite gate compositions from $\mathcal{G} = \{H, T, \text{CNOT}\}$. Objects are finite and discrete – no continuous coordinate, no symplectic form.

2.3 Theorem 1: Non-Exactness

Theorem 1. $F : \mathcal{P} \rightarrow \mathcal{C}$ is not exact – there exist invariants preserved in $\mathcal{P}$ with no image in $\mathcal{C}$.

Proof. By Gromov’s non-squeezing theorem (1985): the symplectic capacity $c_G(M, \omega)$ is invariant under symplectomorphisms. Under F , this maps to a finite discrete point set with zero capacity. An invariant in the source with no image in the target – the defining signature of a non-exact functor. $\square$

3. Algebraization Complexity

The transmon Hamiltonian contains $\cos(\phi)$ – a transcendental function. For computation it must be approximated by a finite polynomial.

Definition. The **Algebraization Complexity** $A(H, \varepsilon)$ is the minimum polynomial degree N preserving eigenvalues to within ε .

Theorem 2. For the transmon with $E_J/E_C \gg 1$:

$$A(H, \varepsilon) \approx \sqrt{\frac{8E_J}{E_C}} \cdot \frac{1}{2} \ln \left(\frac{1}{2\varepsilon} \right)$$

For $E_J/E_C = 50$, $\alpha_r = 1.9\%$, $\varepsilon = 10^{-4}$: $A \approx 224$.

Result: The transmon's exponential charge-noise immunity $\exp(-\sqrt{8E_J/E_C})$ is mathematically identical to its algebraic intractability – a **Coherence-Algebraization Conservation Law**.

4. Thermodynamics of Translation

4.1 Approximation Entropy S_A

The Kullback-Leibler divergence between the full bosonic density operator and its discretized projection:

$$S_A = D_{\text{KL}}(\rho_{\text{full}} \parallel \rho_{\text{grid}}) = \ln \left(\frac{N_{\text{max}}}{d_{\text{comp}}} \right)$$

For the transmon ($N_{\text{max}} \approx 12$, $d_{\text{comp}} = 2$): $S_A \approx 1.79$ nats ≈ 2.58 bits per state projection.

4.2 Rosetta's Constant R

$$R \equiv k_B T \cdot S_A = k_B T \cdot \ln \left(\frac{N_{\text{max}}}{d_{\text{comp}}} \right)$$

At $T = 15$ mK: $R \approx 3.7 \times 10^{-25}$ J per state projection. Per logical gate: $R_{\text{gate}} \approx 1.5 \times 10^{-24}$ J/gate. With all five entropy sources (§7): $Q_{\text{total}} \approx 1.4 \times 10^{-24}$ J/gate.

4.3 Landauer Extension

Landauer (1961) established $k_B T \ln 2$ for erasing one bit within the SAME domain. We extend to inter-domain:

$$Q_{\text{total}} = k_B T \ln 2 + k_B T \ln(N_{\text{max}}/2) = k_B T \ln(N_{\text{max}})$$

For the transmon: $Q_{\text{total}} \approx 2.48 k_B T$ – the Rosetta term is $3.6\times$ the Landauer term.

5. The Digital Speed Limit

5.1 Trotter Error

The standard digital simulation uses: $e^{-i(H_A+H_B)t} \approx (e^{-iH_A t/n} e^{-iH_B t/n})^n$. Error scales as $\|[H_A, H_B]\|$. For the transmon: $\|[H_{\text{harm}}, H_{\text{nonlin}}]\| \propto \hbar \omega_p \alpha_r$.

5.2 Trotter Wall

For fault-tolerant threshold $\varepsilon = 10^{-4}$:

$$\frac{G_{\text{effective}}}{G_{\text{advertised}}} \approx \frac{\omega_p}{\alpha_r} \approx 50$$

A “100-gate” circuit executes 5,000 sub-gates. True gate budget at $T_2^* \approx 100 \mu\text{s}$:

$$G_{\text{budget}} \approx \frac{T_2^*}{t_g \cdot (\omega_p / \alpha_r)} \approx 100$$

5.3 Fractal Boundary

Theorem 4. There exists $\alpha^* \approx 3\%$ below which weakly anharmonic bosonic arrays are classically simulable. Standard transmons at $\alpha_r \approx 1.9\%$ are BELOW this boundary.

6. The Translation Calorimeter

Two experiments on the same transmon at the same temperature:

Experiment	Method	Predicted Heat
A (Analog)	Natural evolution under H_0	$Q_A \approx 0$
B (Digital)	Trotterized gate sequence	$Q_B \approx k_B T \cdot S_A$

Falsification: If $\Delta Q = Q_B - Q_A = 0$ (within noise), the framework is refuted.

Feasibility: NIS junction calorimeter with 1 μK sensitivity. $Q_{\text{single}} \approx 8.2 \times 10^{-25}$ J; 10^6 cycles yield $\Delta T \approx 0.82$ mK (measurable). 37 gates for 5σ (theory). Estimated cost: \$520K, 18–24 months.

7. Hidden Axioms Layer (HAL)

$$S_{\text{Total}} = S_{\text{Obs}} + S_{\text{Base}} + S_{\text{Arch}} + S_{\text{Alg}} + S_{\text{Trot}}$$

Term	Origin	Transmon Value (nats)
S_{Obs}	Human cognitive filter (binary thresholding)	1.79
S_{Base}	Radix mismatch (binary vs. native spectrum)	1.79
S_{Arch}	Archimedean time-slicing (uniform Trotter)	0.5
S_{Alg}	Algebraization (cosine $\rightarrow$ polynomial)	1.0
S_{Trot}	Commutator error ($[H_{\text{harm}}, H_{\text{nonlin}}] \neq 0$)	1.5
Total		6.6

Anthropocentricity: The qubit encoding is a human cognitive artifact. Binary is the radix humans chose because we have ten fingers and built classical computers in base-2. A Platonic observer with access to all d resolvable levels would see a qudit, not a qubit: $S_{\text{Obs}} = \ln(d/2) \approx 1.79$ nats.

Radix Contingency (Strengthened by Planck-Radix Correction): Planck’s quantization gives a **unary/integer basis** ($n = 0, 1, 2, 3, \dots$). This is the natural radix of any harmonic oscillator mode. The field imposes base-2 by truncating to $n \in \{0, 1\}$. The entropy cost is **not** the cost of discretization (the spectrum was already discrete). It is the cost of **radix conversion**: mapping a base- d system to base-2. For $d = 12$: $S_{\text{Base}} = \ln(d/2) \approx 1.79$ nats. This corrects V1.0, which mischaracterized S_{Base} as “continuous $\rightarrow$ discrete translation.” The spectrum was always discrete. The translation is base- $d \rightarrow$ base-2.

Continuum Fallacy: The Trotter error IS the Archimedean cost. Logarithmic time-stepping reduces sub-gates by ~50% with no hardware modifications.

8. Discussion: The Qudit Manifesto

8.1 The Transmon Was Never a Qubit

The core error of the gate-model quantum computing program is not engineering. It is conceptual. Planck gave us a discrete, unbounded integer ladder in 1900. For 125 years, we have built machinery that treats this ladder as if it had exactly two rungs. The “qubit” is a human cognitive artifact – binary is convenient for engineers with ten fingers, not a physical constraint.

The transmon’s 12 resolvable levels (Wang et al. 2022) were never a “leakage problem.” They are the system’s native computational substrate. The leakage is not the system failing at being a qubit. The leakage is the system succeeding at being a bosonic mode – and us misinterpreting its correct behavior as an error.

8.2 Qudit Quantum Error Correction Exists

The constructive alternative is already known. Qudit surface codes (Gottesman 1999; Campbell 2014) generalize the standard qubit surface code to d -dimensional quantum digits. The stabilizer formalism extends naturally:

$$X|j\rangle = |j + 1 \bmod d\rangle, \quad Z|j\rangle = \omega^j|j\rangle$$

where $\omega = e^{2\pi i/d}$. Qudit codes offer:

1. **Higher threshold:** Qudit surface codes have thresholds that increase with d (Campbell 2014)
2. **Reduced overhead:** A single qudit encodes $\log_2(d)$ bits, reducing physical-to-logical qubit ratios
3. **Native hardware match:** The transmon’s 12-level ladder maps directly to a $d = 12$ qudit – no truncation, no leakage

The field should be building qudit surface codes on transmon hardware. The transmon is not a bad qubit. It is a **good qudit** waiting for qudit error correction.

8.3 The Planck Fidelity: A New Benchmark

We propose a benchmark metric: **Planck Fidelity** $\mathcal{F}_P$ – the fidelity of a quantum operation when the occupation number n is mapped to a qudit basis without truncation:

$$\mathcal{F}_P = \text{Tr} \left(P_{\text{qudit}} U_{\text{gate}} |\psi\rangle\langle\psi| U_{\text{gate}}^\dagger \right)$$

where $P_{\text{qudit}} = \sum_{n=0}^{d-1} |n\rangle\langle n|$ (all resolvable levels, not just $\{0, 1\}$). For any finite

gate time, $\mathcal{F}_P > \mathcal{F}_{\text{qubit}}$ because the drive couples to ALL n via $a^\dagger$ – the higher rungs are participating whether we acknowledge them or not.

8.4 Paths Forward

We propose three constructive paths, prioritized:

1. **Qudit QEC on transmons (primary):** Build and benchmark $d = 12$ qudit surface codes on existing transmon hardware. This requires no new fabrication – only a firmware and control software update to address all 12 levels. The existing hardware IS a qudit processor. We just need to program it as one.
2. **Qudit-native algorithms:** Grover’s search, Shor’s factoring, and quantum simulation all have qudit generalizations (Wang et al. 2020) that exploit the larger Hilbert space per physical element. A $d = 12$ qudit encodes $\log_2(12) \approx 3.58$ bits per element – a $3.6\times$ information density improvement over qubits with zero additional hardware.
3. **Hybrid qubit-qudit architectures:** During the transition, run qubit algorithms on the $\{|0\rangle, |1\rangle\}$ subspace while developing qudit QEC in parallel. This is the pragmatic path for existing NISQ-era systems.

8.5 Falsifiability

This framework makes specific, testable predictions: 1. $\Delta Q > 0$ in calorimetry (the binary truncation has a measurable thermodynamic cost) 2. $\Delta Q \propto -\ln(d/2)$ across transmon variants with different d 3. $\alpha^* \approx 3\%$ Fractal Boundary 4. Trotter Wall $\approx 50\times$ overhead for binary gates 5. Planck Fidelity $\mathcal{F}_P > \mathcal{F}_{\text{qubit}}$ for any gate operation on a transmon with $d > 2$

If $\Delta Q = 0$ – if the binary truncation has zero thermodynamic cost – the entropy framework collapses. If $\mathcal{F}_P \leq \mathcal{F}_{\text{qubit}}$ – if including higher rungs does not improve fidelity – the qudit thesis is refuted. This is the hallmark of good science.

9. Conclusion

Planck gave us a digital computer in 1900. We spent 125 years reading it in the wrong base. The transmon is a qudit. The field’s insistence on binary truncation is a radix error, not a physics error. Planck’s quantization gave us a discrete ladder. We chose to climb only the first two rungs – and then spent 35 billion dollars wondering why we kept falling off.

The Approximation Entropy S_A framework developed in this paper quantifies the cost of that choice. The Qudit Manifesto proposes the alternative: stop fighting the transmon’s bosonic nature. Start programming the machine Planck actually gave us.

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